

Qatar-1b

R-1735

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_160516 · created 2026-08-02T18:00:14+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457524.90669 +/- 0.0008 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.107 +/- 0.02
Transit depth (Rp/Rs)^2	1.14 +/- 0.42 [%]
Orbital Inclination (inc)	85.27 +/- 1.95
Airmass coefficient 1 (a1)	1.34 +/- 0.015
Airmass coefficient 2 (a2)	-0.1952 +/- 0.0094
Scatter in the residuals of the lightcurve fit is	1.08 %
Best Comparison Star	None
Optimal Aperture	2.82
Optimal Annulus	5.84
Transit Duration (day)	0.0697 +/- 0.0098

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.107 ± 0.02 vs 0.1463 (deviation 1.66σ)
Duration fit vs published	0.0697 d vs 0.0692 d (deviation 0.05σ)
Mid-time O–C	1.9 min
Depth SNR	4.5
Residual scatter	1.08 %

Light curve

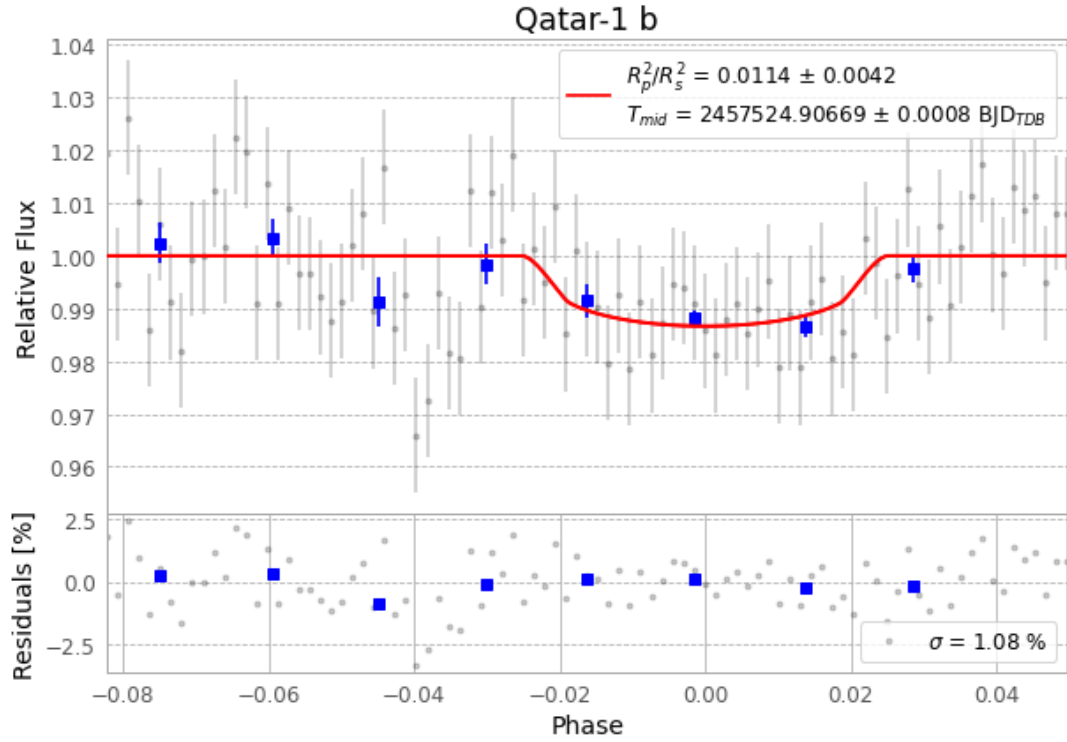


Figure 1 — FinalLightCurve_Qatar-1 b_2016-05-15.png (SHA-256 02fd95ca0af6475e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [312, 274], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8544b0aa776ac127...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

91 FITS frames (60 MB), Qatar-1160516065712.FITS → Qatar-1160516112712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-160516.log (d9a39e2e1a10...), FinalLightCurve_Qatar-1 b_2016-05-15.png (02fd95ca0af6...), FinalLightCurve_Qatar-1 b_2016-05-15.csv (05c361244e5e...)

Compute resources

Wall time 185.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 18:00Z.